

LITERATURE SURVEY

Author & Year	Technology	Key Findings	Limitations
Alzubaidi et al. (2025)	Fusion of Pretrained CNN Models	Proposed a fusion of multiple pretrained CNN models to improve cervical cell classification accuracy. The feature fusion approach extracted complementary information from different deep learning models, resulting in better performance than individual CNNs.	The model was evaluated only on benchmark datasets and lacked validation using real-world clinical data, limiting its practical applicability.
Sharma et al. (2024)	EfficientNetB3 + Attention + Grad-CAM	Developed a cervical cell classification model using EfficientNetB3 integrated with an attention mechanism and Grad-CAM. The approach enhanced feature extraction, improved classification accuracy, and provided explainable predictions by highlighting important image regions.	The study relied on a single deep learning model and did not investigate model fusion techniques that could further improve classification performance.
Plissiti et al. (2018)	SIPaKMeD Dataset with CNN Models	Introduced the SIPaKMeD benchmark Pap smear dataset, which has become one of the most widely used datasets for cervical cell classification research. It enabled researchers to train and evaluate CNN-based classification models effectively.	The dataset includes only cervical cell-type labels and does not provide cervical cancer stage information, restricting its use for cancer stage prediction.